BT4222 Final Report

SPOTLIGHT: A News Recommendation Framework for Homepage Slate Optimisation

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Abstract (Standalone page, within 300 words)

- **Business Objective & RecSys Formulation:** Define the core recommendation problem grounded in an exploratory understanding of dataset distributions and domain constraints.
    - Top K & cold start
- **Prior Baseline & Critical Diagnosis:** Frame your starting point with reference to past project baselines (e.g., NCF, NMF), identifying their structural limitations, methodological gaps, or performance bottlenecks.
- **Group Contributions & Modeling Highlights:** Summarize your group's engineering, architectural, and analytical improvements over the baseline.
- **Evidence-Backed Quantitative Insights:** State key business and behavioral insights supported directly by concrete quantitative metrics, offline evaluation, or diagnostic error analysis → Ablation

1. Data Understanding & Feature Engineering

1.1 Changes to Interim Features Dictionary

Table 1.1a: Kept/Modified Features

Feature	Description	Hybrid UBCF/CBF	MLP	NRMS
article_content_emb **Status:** Kept	**Interim:** 384-d article embedding of the article's semantic content. **Final:** Unchanged. All ll-MiniLM-L6-v2 on title + abstract, L2-normalised. A missing abstract is replaced by the title.	✔	✔	
category_match **Status:** Kept	**Interim:** 1 if the candidate is in the user's most-read category, else 0. **Final:** Unchanged. It is set to 0		✔	

	for empty history.			
reading_entropy **Status:** Kept	**Interim:** Shannon entropy of the user's history category distribution. **Final:** Unchanged. It is set to 0 for empty history.	✔	✔	
has_history **Status:** Kept	**Interim:** 1 if the user has at least one clicked article. **Final:** Unchanged.		✔	
kg_entity_emb **Status:** Modified	**Interim:** Mean of the 100-d TransE vectors of entities in the title and abstract with no missing-value handling **Final:** Mean is L2-normalised. Articles with no mapped entity keep a zero vector.	✔	✔	
category_id / subcategory_id **Status:** Modified	**Interim:** 18 categories (16-d embedding) and 270 subcategories (32-d), from the full catalogue, with separate tables. **Final:** Vocabularies fitted only on training-period articles. Both embeddings are 32-d. One category table is shared by the article and user sides. Unseen categories and subcategories are mapped to UNK.		✔	
historical_ctr **Status:** Modified	**Interim:** Raw CTR from exposures (clicks / exposures) before a global timestamp t. **Final:** Smoothed CTR, (clicks + 20·priors) / (exposures + 20), where priors is the training-period CTR. Counts are lagged and calculated from days strictly before the impression day.	✔	✔	
user_history_emb	**Interim:** Multi-head attention-weighted average of the	✔	✔	

Status: Modified	clicked articles' content embeddings. **Final:** Static L2-normalised mean of the clicked articles' MiniLM vectors. It is set as zero vector for empty history.			
user_category_affinity **Status:** Modified	**Interim:** Normalised 18-d vector of the user's historical category reads. **Final:** Aligned with the training category vocabulary (17 categories plus PAD and UNK, 19 dimensions in total). It is set as a zero vector for empty history.		✔	
log_history_length **Status:** Modified	**Interim:** Raw count of clicked articles in the history. **Final:** log(1 + history_length) to compress heavy right tail to prevent very long histories from dominating. The raw count still defines the cold (0), sparse (1–4) and warm (≥5) evaluation groups.	✔	✔	✔
hour_sin, hour_cos **Status:** Modified	**Interim:** Integer hour (0–23) with a 16-d learned embedding. **Final:** Replaced by sin and cos of $2\pi h/24$ to retain the circular nature of time.	✔	✔	

Table 1.1b: Removed Features

Feature	Description	Reason for removal
article_recency	Elapsed time (in hours) between an article's first appearance in the logs and the current impression timestamp.	MIND does not provide publication timestamps, thus article recency is approximated using each article's first appearance in the logs. Therefore, all articles first appearing on Day 1 receive a recency of 0, while articles published before Day 1 are incorrectly treated as newer than they are. This distortion is the most severe during

		training and least severe on the test day, causing the model to learn a weak and inconsistent relationship between recency and engagement.
session_dow	Day of the week (0 = Monday … 6 = Sunday).	With the chronological 5-1-1 split, the validation and test weekdays never appear in training, hence their embeddings are never trained. hour_sin and hour_cos are used instead to provide time context as they appear in every split.
inactivity_gap	Hours since the user's previous recorded impression.	Inactivity gaps before the first training day cannot be observed. Moreover, as all articles within an impression share the same value, inactivity_gap cannot rank articles within an impression.
slate_length	Number of candidate articles in the impression.	Slate length is the same for every candidate in an impression, so it has no discriminative power to rank them.

Table 1.1c: Added Features

Feature	Description	Hybrid UBCF/CBF	MLP	NRMS
cos_user_hist	Cosine between user_history_emb and the candidate's MiniLM embedding. Empty history is imputed with the training mean.	✔	✔	
user_cold	1 if no clicks are known for the user at scoring time: nothing in the build-window log and an empty history on arrival. Unlike has_history, which looks only at the impression's own history, it also counts clicks from earlier days in the window.	✔		
max_cos_hist	Highest cosine between the candidate and any single past click to capture a narrow interest that averaging might diminish. Empty history is imputed with the training mean.		✔	
cos_last5	Cosine between the candidate article's MiniLM embedding and the mean MiniLM embedding of the user's 5 most recent articles. Empty history is		✔	

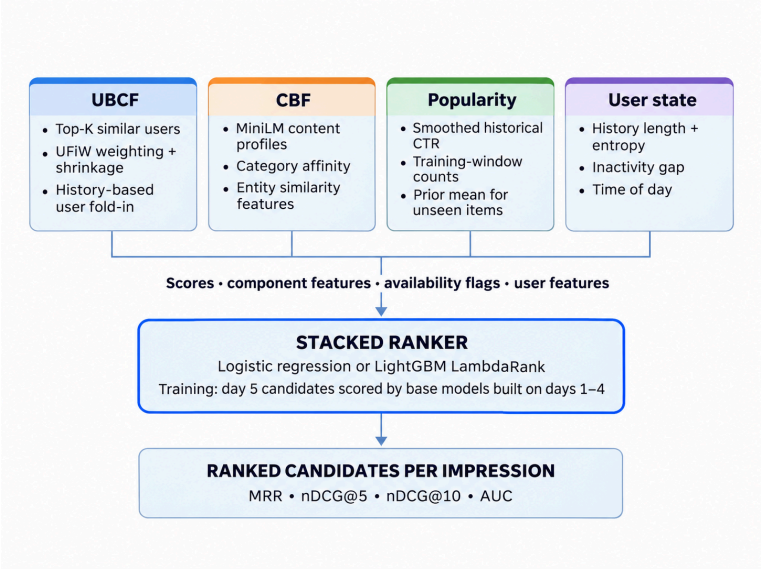
	imputed with the training mean.			
cat_affinity	Share of the user's history clicks in the candidate's category. Empty history is imputed with the training mean.	✔	✔	
kg_match	Cosine between the user's KG profile (mean kg_entity_emb of their clicked articles) and the candidate's kg_entity_emb. Imputed with the training mean when either side has no entity embeddings.	✔	✔	
has_kg_entity	Missingness flag for kg_entity_emb. 1 if the article has at least one mapped TransE entity.		✔	
log_exposure	log(1 + exposures before the impression day). 0 for articles never shown before.		✔	
is_cold_article	Indicates if an article is cold (i.e., received no clicks before the impression day).	✔	✔	
diversity_ratio	Calculates the number of distinct categories in the history / history length. Set to 0 for an empty history.		✔	
habit_score	$\cos(2\pi \cdot \Delta t/24)$, where Δt is the hours since the user's previous impression. Close to 1 when the user returns at the same time of day. A user's first visit is imputed with the training mean.		✔	
click_history_seq	The 50 most recent article indices per user with zero-padding.		✔	✔
title_tokens	Lower-cased title tokens, initialised with 300-d GloVe vectors and fine-tuned. Words outside the vocabulary map to UNK and titles shorter than 30 tokens are padded with PAD (0),			✔
pop_sets	Previous day's category-stratified popular articles (the top 2 of every category with ≥20 clicks, plus 10 slots shared by click volume), with zero-padding. Day 1 uses its own clicks.			✔

ubcf_score, ubcf_strength, ubcf_n_clicks	Similarity-weighted share of the user's 200 nearest neighbours who clicked the candidate, the neighbours' mean similarity, and the number of neighbours who clicked.	✔		
ubcf_available, cbf_available, pop_available	1 if that model can score the pair, else 0.	✔		

2. Models and Performance (Within 4 pages)

2.1 Architecture Diagrams

2.1.1 User-based Collaborative Filtering (UBCF) and Content-based Filtering (CBF)



2.1.2 Multi-Layer Perception (MLP)

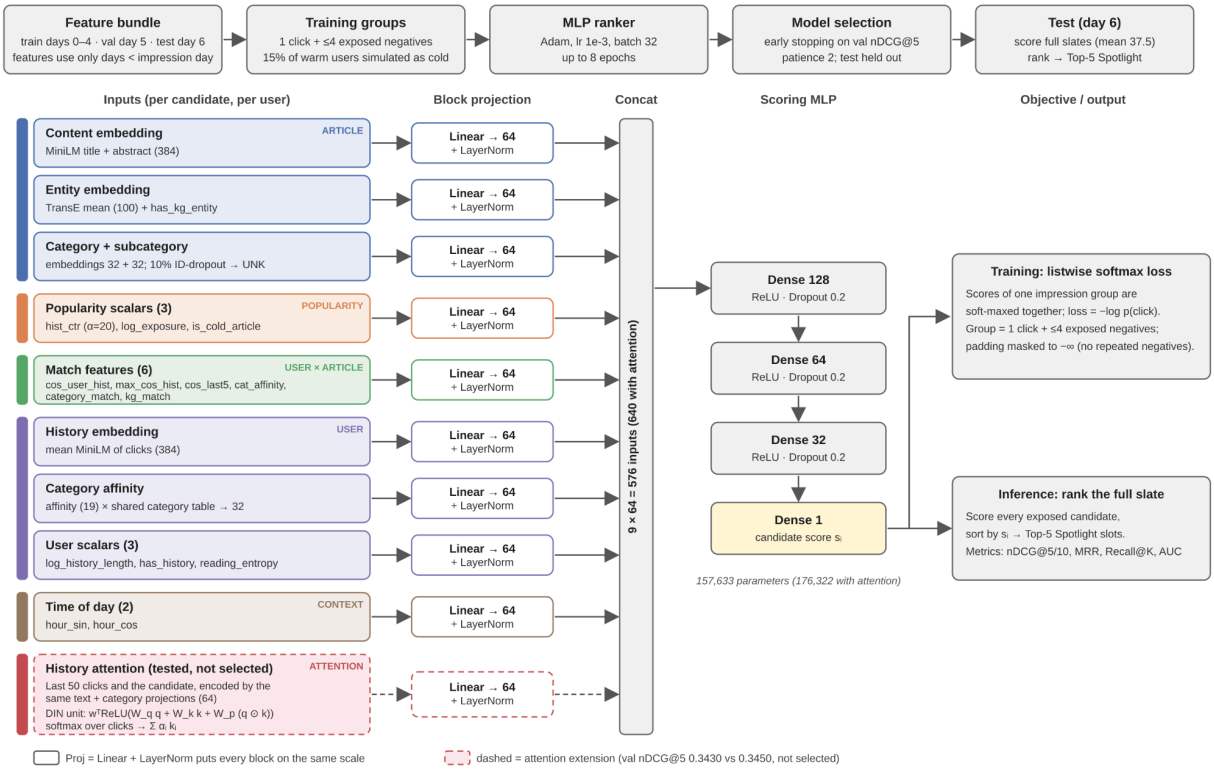


Fig XX: MLP Ranker Architecture

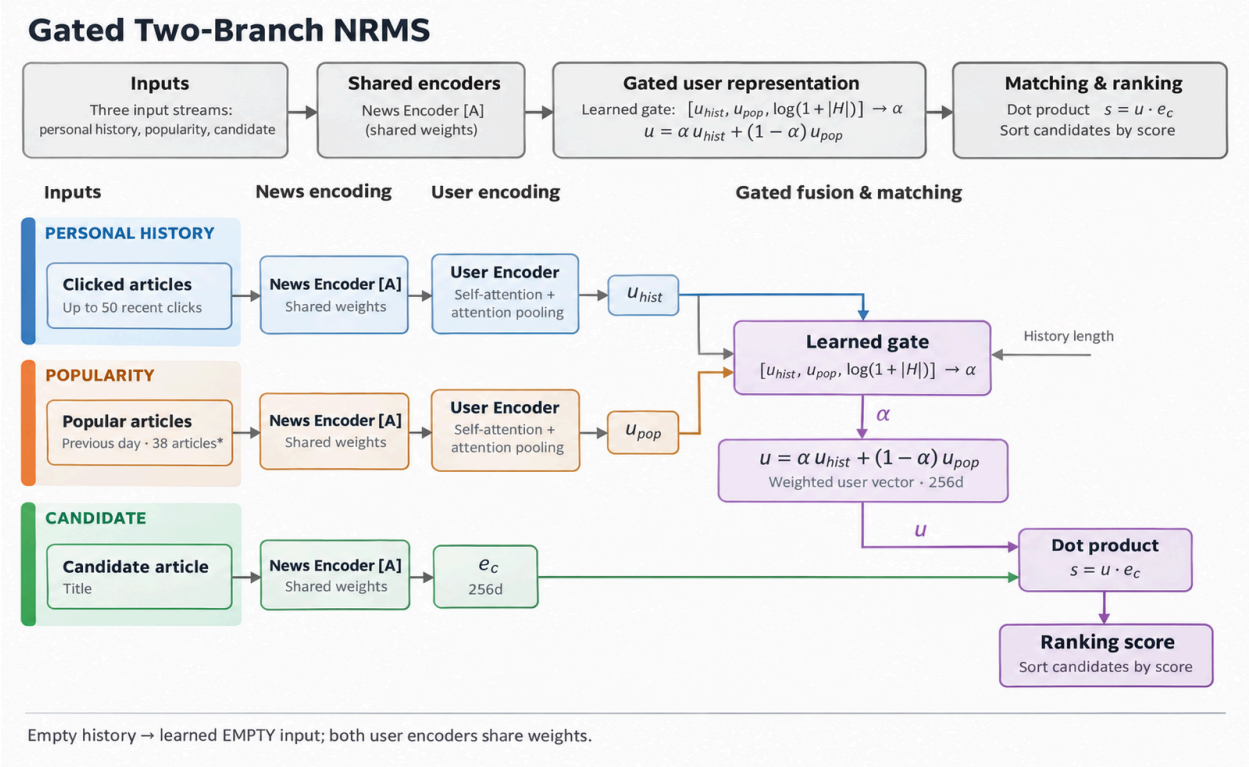
Base MLP: Each candidate article is describe by nine feature blocks: *article content, entities, categories, popularity, user x article match, user history, user category affinity, user scalars* and *time of day*. Each block is projected to 64 dimensions with LayerNorm, so no block dominates. The nine 64-d vectors are concatenated into 576 inputs and passed through three hidden layers (128, 63, 32 units, ReLU, dropout 0.2), which output one score per candidate. A listwise softmax loss over each click and up to four exposed but unlock articles rewards the model for ranking the click first. At test time, the top five scored article fill Spotlight slots

Improved MLP (final model): The Base MLP lost to popularity on cold users, because only 1.9% of training examples had no history. In the improved MLP, 15% of users with history in each training batch are shown as if they had none, which teaches the model how to rank for new users. Ablation also dropped diversity_ratio and habit_score.

Alternative: history attention (in red on Fig X). Given that the Improved MLP uses one average of the user history for every candidate, an alternative MLP model was proposed to weight each past click by its relevance to the candidate scored. One averaged history vector blurs the interest of users who read across several categories. Following the Deep Interest Network (Zhou et al., 2017), we tested an attention block that weights each of the user's last 50 clicks by relevance to the candidate, added as a tenth 64-d block. This helps readers with mixed interests, whose single averaged profile blurs several categories together.

However, as the model scored lower on validation than the original nine block MLP, the attention block was removed for the final model.

2.1.3 Neural News Recommendation with Multi-Head Self-Attention (NRMS)



NRMS baseline. NRMS (News Recommender Neural News Recommendation with Multi-Head Self-Attention) is a personalized, content-based neural recommendation model. It represents each news article from its title and each user from the articles they have clicked. The News Encoder first maps 300-dimensional GloVe word embeddings to 256 dimensions, then applies multi-head self-attention to model relationships between title words. Additive attention pooling combines the word representations into one 256-dimensional news vector. The same News Encoder processes up to 50 clicked articles in a user's history. The User Encoder applies self-attention across those article vectors, then attention pooling combines them into a 256-dimensional user-interest vector.

Gated NRMS. We added a popularity branch alongside the personal-history branch and used a learned gate to combine their signals. The popularity branch selects articles from the previous day's clicks: eligible categories must have at least 20 clicks, each contributes its two most-clicked articles, and 10 additional slots are allocated by click share. In the reported data, 14 eligible categories yielded 38 articles.

Gated NRMS improves on standard NRMS overall, with the clearest gains for cold users: nDCG@5 increases from 0.2384 to 0.2675, MRR from 0.2582 to 0.2852, and AUC from 0.4875 to 0.5520. Sparse-user results are mixed, with nDCG@5 unchanged and nDCG@10 slightly lower. The gate's α for sparse users averaged 0.638 ± 0.161 across three seeds (i.e 42, 43, 44,...), indicating substantial variation in how the two branches were weighted; this instability may help explain the limited improvement for that group.

2.2 Models Performance

All users include warm, sparse and cold users. Warm users refer to users with history length of 5 or more. Sparse users refer to users with history length of 1 to 4. Cold users refer to users with history length of 0.

2.2.1 All Users

Validation Performance

Test Performance

Model	nDCG@5	nDCG@10	MRR	AUC
Popularity				
UBCF				
CBF	0.3461	0.4042		
Improved MLP	0.3567	0.4193	0.3253	0.6667
Attention MLP	0.3599	0.4228	0.3281	0.6714
NRMS	0.3145	0.3792	0.3320	0.6257
Gated NRMS	0.3198	0.3834	0.3361	0.6325

*Attention MLP was not selected as its validation nDCG was lower than Improved MLP nDCG

2.2.2 Warm Users

Validation Performance

Test Performance

Model	nDCG@5	nDCG@10	MRR	AUC
Popularity				
UBCF/CBF				
Improved MLP	0.3598	0.4225	0.3267	0.6765
Attention MLP	0.3628			
NRMS	0.3138	0.3791	0.3335	0.6345
Gated NRMS	0.3190	0.3831	0.3375	0.6398

2.2.3 Sparse Users

Validation Performance

Test Performance

Model	nDCG@5	nDCG@10	MRR	AUC
Popularity				
UBCF/CBF				
Improved MLP	0.3552	0.4175	0.3340	0.6185
Attention MLP	0.3613			
NRMS	0.3404	0.4013	0.3407	0.5949
Gated NRMS	0.3404	0.4006	0.3390	0.5979

2.2.4 Cold Users

Model	nDCG@5	nDCG@10	MRR	AUC
Popularity				
UBCF/CBF				
Improved MLP	0.2752	0.3336	0.2536	0.5631
Attention MLP				
NRMS	0.2384	0.3004	0.2582	0.4875
Gated NRMS	0.2675	0.3269	0.2852	0.5520

3. Contribution and Justification (1–3 pages per chosen aspect)

Complete the Self-Evaluation Table below. Groups are not required to cover all optional aspects; focus on the areas matching your team's technical depth.

<table>
<tr><th>Contribution Aspect</th><th>Dimension</th><th>Basic</th><th>Optional / Advanced</th></tr>
<tr><td rowspan="2">1. Data Understanding, Rich Signal Utilization & Feature Engineering
(Impressions, multi-type user interactions, context, domain features)</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
<tr><td colspan="4">Clarification:
● Basic: e.g. Convert data into standard binary interaction pairs without deeper exploration of additional information; apply standard transformations or descriptive statistics.
● Optional: e.g. Thoroughly analyze data distributions and sparsity prior to modeling; exploit rich signals (e.g., impression logs, exposed vs. unexposed items, multi-behavior interactions, timestamps/sessions) and engineer domain-informed features to enrich predictive signals.</td></tr>
<tr><td rowspan="2">2. Prior Baseline Benchmarking & Gap Diagnosis

(Critical evaluation of NCF, NMF, and conventional baselines)</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
<tr><td colspan="4">Clarification:
● Basic: e.g. Replicate standard baseline models (e.g., NCF, NMF) directly to obtain comparative metrics.
● Optional: e.g. Critically diagnose structural failure modes or information bottlenecks in baseline implementations; systematically justify your improvements against these identified gaps.</td></tr>
<tr><td rowspan="2">3. Designing or Adapting Recommender Architectures</td><td>Effort</td><td></td><td></td></tr>
<tr><td>Effectiveness</td><td></td><td></td></tr>
<tr><td colspan="4">Clarification:
● Basic: e.g. Direct implementation of standard models or simple ensembling.
● Optional: e.g. Adapt or customize architectures capturing temporal dynamics, sequential interaction patterns, or network/relational structures to improve recommendation quality.</td></tr>
<tr><td rowspan="2"><s>4. Model Diagnostics, Error Analysis & Quantitative Insights</s></td><td><s>Effort</s></td><td></td><td></td></tr>
<tr><td><s>Effectiveness</s></td><td></td><td></td></tr>
<tr><td colspan="4"><s>Clarification:
● Basic: e.g. Describe aggregate test metrics without deeper sub-group analysis
● Optional: e.g. Perform granular error diagnostics across distinct user/item segments; conduct deep learning model interpretation by examining learned parameters (e.g., analyzing attention</s></td></tr>
</table>

~~weights, etc.); extract actionable behavioral and business insights supported by empirical evidence~~			
~~5. Smart Use of AI Tools & Critical Collaboration~~	**~~Effort~~**		
	~~Effectiveness~~		
~~*Clarification:*~~ • **~~Basic:~~** ~~Use AI tools strictly for routine coding syntax, formatting, or text polishing~~ • **~~Optional:~~** ~~Engage AI as a critical sparring partner to explore modeling alternatives, catch conceptual/coding flaws, validate technical logic, and drive expanded learning documented in the AI Interaction Log.~~			

3.1 Data Understanding, Rich Signal Utilisation & Feature Engineering

All three models draw on a shared set of findings from the EDA. F1–F6 describe the data's distributions, and F7–F10 describe its feedback and logging properties. Sections 3.1.1–3.1.3 explain how each model's features respond to them.

Distributional findings

1. **F1: Extreme popularity skew.** 1% of articles drive 54.3% of clicks, and 60% of articles have a CTR of 0.
2. **F2: Most candidates are cold articles.** More than 50% of candidate articles have no clicks before their impression day.
3. **F3: Users are highly selective.** The mean slate has 37.3 articles, yet 72.1% of impressions have exactly one click.
4. **F4: User history is skewed and often missing**. History length is heavily right-skewed: the median is 12 clicks, the mean is 21.7, and the maximum of 558 is over 20 times the upper quartile of 26. 17.91% of users have fewer than 5 clicks and 2.26% have none.
5. **F5: Time-of-day patterns recur.** Reading activity follows the same daily cycle across the 7-day window, peaking at around 8 to 9 am.
6. **F6: Users read across several categories.** Mean reading entropy is 1.43, equivalent to about 4 equally read categories.

Feedback and logging findings

7. **F7: History has no timestamps.** The history field records past clicks without times, and some may predate the 7-day window. Only impression clicks are timestamped.
8. **F8: Feedback has three states.** An article can be clicked, shown but not clicked, or never shown. Never-shown articles are unobserved, not negative. Even unclicked labels are weak: 7.06% of repeatedly shown user–article pairs were clicked in one impression and ignored in another.

9. **F9: Popularity reflects exposure.** Because 60% of exposed articles were never clicked, a low click count may mean an article was rarely shown rather than disliked.
10. **F10: Behaviour is stable across splits.** CTR is 4.1%, 3.8% and 4.1% across train, validation and test, and the hourly activity pattern is consistent, so later days are representative of earlier ones.

3.1.1 UBCF/CBF

The hybrid's 16 stacker features are grouped into four blocks: collaborative, content, popularity and user state. Their design draws on F1, F2, F4 and F6–F10 (Section 3.1).

Collaborative features are engineered to separate weak from well-supported neighbour evidence (F2, F4, F7, F8). The UBCF user-article matrix combines reading histories with impression clicks and records positives only, since unobserved articles carry no negative evidence. Each user's history on the scoring day is added to their profile, for users missing from the training window to be matched to neighbours. Because ubcf_score is a ratio, a sparse user whose single neighbour clicked receives a higher score than a warm user for whom 50 of 200 neighbours clicked. ubcf_n_clicks (the number of neighbours who clicked) and ubcf_strength (the neighbours' mean similarity) therefore indicate how much evidence supports each score.Time decay was tested but rejected, because history clicks have no timestamps, and decay lowered validation nDCG@10 in every configuration. As 56.9% of validation and 81.9% of test candidates have no prior clicks, UBCF scores only 42.9% and 17.6% of pairs. ubcf_available therefore stops the stacker from mistaking missing evidence for low relevance.

Content features are engineered to score articles without prior clicks (F2, F4, F6). cbf_content (MiniLM), cbf_kg (TransE) and cbf_cat_aff compare each candidate with the user's content, entity and category profiles. They do not depend on the article having been clicked, so CBF scores 99.5% of validation and 97.3% of test pairs. The three signals are passed to the stacker separately rather than as one fixed-weight score. Validation tuning set the KG weight to zero in the standalone CBF score, yet cbf_kg accounts for 5.2% of LightGBM's gain, showing that the stacker uses entity overlap in ways a single linear weight cannot. Articles without mapped entities keep a zero KG vector.

Popularity features are engineered to differentiate unpopular from unexposed articles (F1, F2, F9). As 60% of exposed articles were never clicked, a raw click count cannot distinguish rarely shown articles from those that were shown often and ignored. pop_ctr is therefore the smoothed CTR, (clicks + 20·prior) / (exposures + 20), so unseen articles receive the training prior rather than zero. On test it outperforms raw click counts under the same evaluator (nDCG@10 of 0.344 vs 0.279). item_cold marks articles whose pop_ctr is only the prior. All counts use days strictly before the impression day, and log_exposure is excluded because prior exposure partly acts as a proxy for article recency.

User-state features are engineered to condition the base-model scores on user type (F4, F5, F6). user_cold, log_history_length, reading_entropy, hour_sin and hour_cos are constant within an impression, so they cannot rank candidates on their own; in the logistic stacker, which has no interaction terms, they only shift predicted probabilities. LightGBM can combine them with the base-model scores, for example by splitting on history length before content similarity, and so weight the base models differently for different users. History length is log-transformed so that the long tail, up to 558 clicks, does not dominate.

Training rows are engineered to prevent leakage (F8, F10). The base models are fitted on days 1–4 and scored on day 5, and the stacker learns from these out-of-time scores. It therefore trains on predictions for unseen data, as at test time, and stable behaviour across splits supports treating day 5 as representative of the held-out days. Each click is paired with up to four unclicked articles from the same impression, and unobserved articles are never used as negatives. Evaluation always uses full slates.

3.1.2 MLP

The MLP features are grouped into six blocks: popularity, article, user, user × article, context and attention. Each block is projected to 64 dimensions with LayerNorm, so no block dominates by scale. All MLP blocks are based on findings F1–F6 (Section 3.1).

Popularity features are engineered to differentiate unpopular from unexposed articles (F1, F2). As most articles have a CTR of 0, raw CTR cannot distinguish genuinely unpopular articles from those with limited exposure. historical_ctr is therefore smoothed towards the training prior, with log_exposure and is_cold_article indicating its reliability. log_exposure measures the amount of prior exposure supporting historical_ctr as low values cause historical_ctr to remain close to the training prior while high values more closely reflect the article's real click rate. is_cold_article identifies articles with no prior clicks. All counts are computed using only data from days before the impression day to prevent data leakage.

Article features are engineered to rank articles independent of clicks, supporting both warm and cold articles (F2). As popularity signals are unavailable for cold-start articles, the article block uses metadata, like MiniLM title and abstract embeddings, TransE entity embeddings, and category and subcategory embeddings. For the 12.97% of articles without entity embeddings, a has_kg_entity = 0 flag prevents their zero vector from being misinterpreted as a weak signal. One category table is shared by the article and user sides, placing the user's category affinity and the candidate's category within the same embedding space. 10% of category and subcategory IDs are also randomly masked as UNK during training to allow the MLP model to learn the UNK embedding. The UNK embedding can represent categories first seen in validation or test.

User x Article features are engineered to identify the article each user is most likely to click (F3). Each slate contains multiple articles but typically receives only one click, hence the model needs user–article interaction features for accurate ranking. Matches between users and articles are computed across three dimensions: time, category and entity. Temporal matching captures the user's overall history (cos_user_hist), closest past click (max_cos_hist) and recent interests (cos_last5). Category matching uses cat_affinity and category_match, while entity matching uses kg_match. Ablation results show that removing this block produces the largest drop in nDCG@5, showing its importance for ranking performance.

User features are engineered to handle short and empty histories (F4). History length is transformed as log(1 + x) to compress its long tail and indicate the reliability of history-based features, like user_history_emb, reading_entropy and user–article match features. These features are less reliable for users with few clicks. reading_entropy further indicates how much weight to assign to category signals such as cat_affinity, as focused readers provide stronger category signals and vice versa. Since missing

match features are imputed with training means, has_history flags users with empty history, allowing the model to distinguish missing values from genuine average matches.

Context features are engineered to capture recurring time-of-day patterns (F3, F5). Hour of day is encoded using sine and cosine transformations, allowing the model to learn recurring daily patterns while retaining cyclic nature of time. Four interim features were intended to provide additional context about the interaction, article and user, however they could instead introduce misleading signals under the chronological 5-1-1 split, therefore these features were removed during feature engineering (refer to Table 1.1b for detailed justification).

Attention features are engineered to match each candidate to the user's relevant interests (F3, F4, F6). Users read across several categories (F6), hence a single mean profile blurs their interests and gives the same user vector for every candidate. However, picking the one clicked article from a slate of multiple articles (F3) depends on which part of the history matters for that candidate. The attention block therefore takes each user's 50 most recent clicked articles, zero-padded, and weighs each click by its relevance to the candidate. Each click is encoded with the same text and category projections as the candidate, so both sit in the same space. The 50-click cap bounds the cost despite the long tail (F4, maximum 558) while covering most histories in full (median 12, upper quartile 26). An empty history gives an empty click list, which the block also learns to handle through cold-user simulation.

All features are engineered to avoid leakage under the chronological split. As validation and test data fall on later days, the vocabularies, CTR prior, imputation means and standardisation statistics are fitted on training days only.

3.1.3 NRMS

BECAUSE WE JUST USE ONLY TITLE SO THERE IS LITERALLY NOTHING TO INCLUDE HERE

3.2 Prior Baseline Benchmarking & Gap Diagnosis ***(Based on the previous project)***

*Talk about limitations of past projects

3.2.1 UBCF/CBF

Gaps in past year project	**Fixes in new model**
Users without history cannot be served. UBCF only scores users present in its training matrix and skipped all others. CBF assigns an all-zero content profile when no valid reading history is available.	UBCF adds the history a user brings on the scoring day, so users absent from the training window can still be matched to neighbours. Smoothed popularity in the stack gives a signal when neither base model has evidence.
No hybrid model. The notebooks contain no hybrid scoring function. UBCF outputs	A LightGBM ranker combines UBCF, CBF and popularity, trained on day-5 predictions from base

similarity-weighted interaction shares and CBF outputs logistic regression probabilities, so their values are not directly comparable.	models fitted only on days 1–4.
The two models cannot be compared or combined fairly. UBCF excluded every impression with an unseen user or article, dropping 67.7% of validation and 90.4% of test impressions, while CBF evaluated all of them. Score ties were broken in favour of the clicked article, and the 4-1-2 split left 15 November unused.	All models are scored on the same impressions and full slates, with a shared evaluator that breaks ties randomly with a fixed seed, under a 5-1-1 split.
UBCF similarity is unweighted, untuned and gives no sense of confidence. Widely read articles dominate user similarity, nothing was tuned, and because the score is a ratio, a sparse user whose single neighbour clicked outscores a user for whom 50 of 200 neighbours clicked.	Widely read articles are down-weighted, similarities based on little overlap are shrunk, and k, shrinkage and decay are tuned on validation.
Coarse content representation. CBF represents articles with 50-d topic vectors.	Articles are represented with 384-d MiniLM title + abstract embeddings, with a separate TransE entity profile passed to the stacker as its own feature.
CBF was not tuned. Its four signals were combined by a default logistic regression, with no search over how much each should count.	Time decay τ, category weight β and KG weight γ are tuned on validation, and the stacker then learns how to combine the components.

3.2.2 MLP

Gaps in past year project	**Fixes in new model**	**Evaluation of fixes**
Lack of article content representation. User features were limited to history length, two category ratios, and category and cluster IDs, while articles had BERT, topic and entity embeddings. There is only one knowledge-graph match score feature that connected a user to an article, hence user–article relationships could only form inside the hidden layers. This forces the model to infer user-article relationship and matches implicitly.	**Initial and Improved MLP.** The new MLP introduces user content profile feature through user_history_emb. There are also five explicit user–article match features, which are kg_match, max_cos_hist, cos_user_hist, cos_last5, and category_match. The new kg_match replaces the original knowledge-graph match score from the previous project. These features provide the model with direct signals of	Removing the user × article block causes the largest validation drop among all feature blocks (−0.0246 nDCG@5), confirming that the explicit match signals are the model's most valuable inputs. Their benefit is strongest when personalisation is possible. The MLP's improvement over the popularity model increases with user history length, from −0.008 for cold users to +0.022 for sparse users and +0.083 for

	users' content preferences and how relevant each candidate article is to the user history.	warm users. This suggests that the user-article features are more effective for users with richer interaction histories.
Uneven input scales. Each feature block passed through its own Linear + ReLU layer with no normalisation, causing the scale differences to create imbalanced representations. This likely contributed to CTR dominating the attribution results.	**Initial and Improved MLP.** Each block is projected to 64-d and passed through LayerNorm, which places every block on the same scale before concatenation to prevent any block from dominating by scale.	Ablation results suggest that no single feature block dominates the model, as removing any block leads to a decline in validation nDCG@5. The removal of the popularity block reduces nDCG@5 by only 0.011, compared with the larger 0.0241 decline when CTR was removed in the past-year project. This suggests that LayerNorm helped to prevent the popularity features from disproportionately influencing the model.
Unseen categories embeddings are not trained. Categories not seen in training are mapped to an out-of-vocabulary index that never appears during training. With no ID dropout, its embedding is never learnt and remains at random initialisation.	**Initial and Improved MLP.** During training, 10% of category and subcategory IDs are randomly masked as UNK, allowing the UNK embedding to be learned. This UNK embedding can provide a representation of categories first seen in validation, testing or deployment.	This fix is not evaluated separately because only 1 article maps to the UNK category and 9 to the UNK subcategory, providing too few observations to measure its effect reliably. This fix is still retained to improve robustness for unseen categories in deployment.
The user representation is static. The same user vector is used to score every candidate article, regardless of the candidate's content.	**Attention MLP.** Before scoring each candidate, an attention mechanism reviews the user's 50 most recent clicked articles and assigns greater weight to articles that are most similar to the candidate. This allows for a user representation that is specific to the candidate.	Compared to the Improved MLP, adding attention increases overall test nDCG@5 by 0.0031. The improvement is greatest for sparse users (+0.0061) as their shorter histories provide focused and relevant signals. It is smaller for warm users (+0.0030) as longer histories may introduce noise. The attention MLP performs worse for cold users (−0.0103) as there is no history for the attention mechanism to use.

3.2.3 NRMS

Gaps in past year project	Fixes in new model
The user vector is built only from clicked articles, so it may miss popular topics outside the user's reading history.	A popularity branch encodes articles selected from the previous day's clicks, adding a signal for current interests.
Users with no history share one learned default embedding. The model can still rank candidates, but this user vector cannot reflect recent trends.	The model combines a learned empty-history representation with the daily popularity vector.
The model has no explicit way to adjust the balance between personal history and popularity as history length changes.	A learned gate weights the history and popularity vectors using both vectors and history length.

3.3 Designing or Adapting Recommender Architectures *(Inspiration and result or sth to show that additional element help improve performance)*

All results use the chronological 5-1-1 split, full slates and a shared evaluator. Design choices were made on validation; test was used for final reporting only.

3.3.1 UBCF/CBF

The hybrid was developed in four stages from the previous year's models.

Stage	Gap addressed	Change made	Val nDCG@5	Test nDCG@5
0. Baselines	Past models used different evaluation rules and could not be compared.	Reimplemented past UBCF and CBF under the shared protocol.	UBCF 0.2150 CBF 0.3056	0.2229 0.2370
1. Improved UBCF	Popular articles inflate similarity, overlap of one article looks strong, and	Added inverse user frequency weighting, similarity shrinkage (λ = 10) and fold-in of the user's history on scoring day. Tuned k = 200. Time decay was rejected on validation.	0.2276	0.2300

	unseen users are skipped.			
1. Improved CBF	50-d topic vectors are coarse and weights were never tuned.	Used 384-d MiniLM profiles with category affinity ($\beta = 0.25$). KG weight tuned to $\gamma = 0$ and time decay rejected for the standalone score.	0.3242	0.3461
2. Stack without popularity	No past hybrid existed, and raw UBCF and CBF scores are not on the same scale.	LightGBM ranker over UBCF, CBF and user-state features, trained on day-5 predictions from base models fitted on days 1–4.	0.3259	0.3419
3. Add popularity (final)	Neither base model has evidence for some user–article pairs.	Added smoothed CTR, item_cold and pop_available, then refitted the stack.	**0.3347**	**0.3545**

Stage 3 was selected because it had the highest validation nDCG@5, +0.0148 over CBF. In the validation ablation, removing CBF cut nDCG@10 from 0.3950 to 0.3045, and removing popularity cut it to 0.3833 (cold users: 0.3409 to 0.3259). Removing UBCF changed it only slightly (0.3941). The gain narrowed on test: nDCG@5 matched CBF (0.3450 vs 0.3452), while nDCG@10 (+0.0016), MRR (+0.0019) and AUC (+0.0066) improved slightly. A likely cause is that the clicked article had no prior clicks in 88.9% of test impressions against 72.7% on validation, leaving popularity less to contribute.

3.3.2 MLP

The first stage closes the design gaps in the past project's MLP (Section 3.2.2). Each later stage targets the weakness found in the previous stage's slice analysis.

Stage	Weakness addressed	Architectural change	Result	Decision
1. Initial MLP	Past MLP had no direct user–article signals and uneven input scales (Section 3.2.2).	Five feature groups projected into 64-d LayerNorm blocks, then 128 → 64 → 32, with listwise softmax over each click and four sampled negatives.	Overall 0.3583 vs popularity 0.2834. Cold articles 0.3402 vs 0.1444. Cold users 0.2547 vs 0.2828 (random 0.2375).	Kept as the base. Cold users need a fix.

2. Improved MLP (final)	Cold users are 1.91% of training and all share identical user inputs, so the model applies weights learned from warm users and never learns to fall back on popularity.	Cold-user simulation: in each batch, 15% of users with history are shown as cold while their clicks stay as labels, raising the cold share to about 16.6%. Validation and test are unchanged. Ablation dropped habit_score and diversity_ratio (val 0.3450 vs 0.3440).	Cold users 0.2547 → 0.2752, of which simulation adds +0.0034. Gap to popularity −0.0281 → −0.0076 (95% CI [−0.0193, +0.0047]). Val cold users 0.2644 vs popularity 0.2435. Cost: −0.0016 overall, −0.003 warm.	Selected. No popularity fallback is needed.
3. Attention MLP	One mean vector per user, the same for every candidate, blurs mixed interests (mean reading_entropy 1.43).	Following Deep Interest Network [Zhou et al., 2017], each of the last 50 clicks is scored against the candidate. The weighted sum becomes a tenth block (576 → 640 inputs; 157,633 → 176,322 parameters).	Overall +0.0031 (95% CI [+0.0021, +0.0041]). Sparse +0.0061, warm +0.0030, cold −0.0041 (not significant). Gain by entropy: low +0.0013, medium +0.0037, high +0.0040.	Not selected. See justification below.

Why the Improved MLP is final. Models were selected on validation only, and the Improved MLP scored higher there (nDCG@5 [Improved] vs [Attention]). Choosing Attention because of its test result would use test data for model selection. Its test gain supports the mixed-interest hypothesis, but the confidence interval covers test sampling only, not training randomness. Retraining the same Improved MLP in a different environment moved test nDCG@5 by 0.0094 (0.3600 vs 0.3506), about three times the attention gain. Attention also adds 18,689 parameters (+11.9%) and gives no benefit to cold users.

3.3.3 NRMS

We started from a title-based NRMS and added a popularity branch, combining the two user vectors with a learned gate:

$$\textbf{Score}(c) = \alpha \times (u_{hist} \cdot e_c) + (1 - \alpha) \times (u_{pop} \cdot e_c)$$

Aspect	Past year's NRMS	This project	Purpose
Article representation	BERT title and abstract features, category information and entity representations.	Trainable 300-d GloVe vectors projected to 256-d, 16-head self-attention and additive attention pooling.	One 256-d vector per title, shared by history, popular and candidate articles.
User representation	Encodes clicked articles only.	History branch (up to 50 clicks) gives u_hist. Popularity branch with the same encoders gives u_pop.	Adds a current-interest signal outside the user's own reading history.
Popular set	None.	From the previous day's clicks: top 2 articles of each category with ≥20 clicks, plus 10 slots by click share (38 articles per day).	Popularity without using same-day clicks.
Balancing the two signals	None.	Learned gate α from both vectors and history length.	Adjusts the balance as history grows.
Empty history	One learned default embedding.	Learned EMPTY input combined with u_pop.	Cold users still receive a daily signal.
Candidate scoring	MLP with user, article, topic and context features.	Dot product of the candidate vector with each user vector, weighted by α.	Keeps scoring simple and makes the gate interpretable.

Training used one click and four non-clicked articles from the same impression, sampled softmax cross-entropy, Adam (learning rate 10^{-4}), batch size 128, three epochs and encoder dropout 0.2. History dropout and truncation (each $p = 0.1$) expose the gate to empty and short histories. The best checkpoint was selected by validation nDCG@5.

Users	Model	nDCG@5	nDCG@10	MRR	AUC
All, validation (n = 30,270)	NRMS	0.2986	0.3604	0.3238	0.6358
	Gated NRMS	0.2957	0.3578	0.3209	0.6350

All, test (n = 73,152)	NRMS	0.3223	0.3853	0.3394	0.6327
	Gated NRMS	0.3228	0.3853	0.3391	0.6327
Cold, test (0 clicks, n = 2,214)	NRMS	0.2243	0.2885	0.2447	0.5000
	NRMS + hard fallback	0.2721	0.3299	0.2873	0.5530
	Gated NRMS	**0.2797**	**0.3360**	**0.2960**	**0.5590**
Sparse, test (1–5 clicks, n = 10,768)	NRMS	0.3514	0.4107	0.3512	0.6081
	Gated NRMS	0.3456	0.4046	0.3438	0.6031

The gate's clearest benefit is for cold users: test nDCG@5 rises by +0.0554 over NRMS (95% CI [+0.0440, +0.0673]) and by +0.0077 over a hard popularity fallback (95% CI [+0.0017, +0.0131]). Mean α rises with history, from 0.41 for cold users to 0.52 for sparse users and 0.75 above 30 clicks, so the model relies more on popularity when history is short. The overall test change is within noise (+0.0005, 95% CI [−0.0004, +0.0013]), and sparse users and overall validation are slightly lower. Cold-start support is therefore the demonstrated benefit, not a gain for every user group. The past year's model is not included because its architecture and evaluation setup differ.

4. AI Interaction Log (Appendix)

Provide a concise table documenting key collaborative interactions with AI tools:
1. Core Prompt & Task Objective: The specific question, modeling concept, or technical problem being explored.
2. AI Output Summary: The substantive response or code generated (do not include all of the code generated by AI in the Appendix).
3. Critical Evaluation: Your assessment of limitations, naive assumptions, or errors in the AI response.
4. Refinement & Final Action: How you validated, adjusted, and incorporated the solution into your project

Core Prompt & Task Objective	**AI Output Summary**	**Critical Evaluation**	**Refinement & Final Action**

Explore suitable neural architectures for news recommendation.	Suggested NRMS with attention-based news and user encoders.	NRMS supports content-based ranking but remains dependent on user history for personalization.	Implemented NRMS and evaluated its performance on MIND-small.
Investigate NRMS performance for users without reading histories.	Suggested popularity-based fallback representations.	Popularity signals may introduce bias and cannot capture individual preferences.	Implemented PG-NRMS using the top 20 clicked training articles and evaluated cold-start performance.
Evaluate NRMS architecture and ranking robustness.	Suggested multi-seed evaluation and attention ablation.	Aggregate metrics alone cannot reveal subgroup-specific limitations.	Conducted three-seed evaluation, self-attention ablation, and cold-start analysis.

5. References (Within 1 page)

Comprehensive citations for all datasets, baseline repositories, and referenced academic literature.

[1] Wu, C., Wu, F., Ge, S., Qi, T., Huang, Y., & Xie, X. (2019). Neural News Recommendation with Multi-Head Self-Attention. **Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)**, 6389–6394.

Zhou, G., Zhu, X., Song, C., Fan, Y., Zhu, H., Ma, X., Yan, Y., Jin, J., Li, H., & Gai, K. (2017). Deep Interest Network for Click-Through Rate Prediction. *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*.